Dhruvi Joshi

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EDUCATION

Boston University | GPA 3.92 Boston, MA
Masters of Science, Bioinformatics 05/25

University of Washington, Seattle | GPA 3.61 Seattle, WA
Bachelors of Science, Biomedical Engineering 06/24
Bachelors of Science , Applied & Computational Mathematical Science

SKILLS

Technical: Python (Pandas, Flask, Scikit learn) , R (ggplot2, tidyverse, Deseq2), R shiny, Bash, Command Line, SQL, Containerization (Singularity), Version Control (Git), Workflow management (Nextflow), High-Performance Computing (HPC), HTML, Javascript

EXPERIENCE

Computational Research Intern Boston, MA
Boston University, Dr. Jinying Chen 05/25 - Present

- Evaluating hierarchical clustering, network clustering, LASSO, decision trees, and random forests to select the most representative feature sets
- Creating logistic regression and tree-based generalizable and interpretable Alzheimers risk prediction models
- Preprocessing, cleaning, and feature selecting from biomarker, clinical, social, and demographic datasets

Bioinformatics Research Assistant Seattle, WA
Armbrust Biological Oceanography Lab 03/23 - 08/24

- Quantified and identified transporter proteins from North Pacific environmental sequence data, through shell scripting, python, containerization, and version control
- Designed a bioinformatic workflow to determine viral response in cyanobacteria; drew out 6 up-regulated putative defense genes for further study
- Created informational visuals with the Seaborn library to present trends for lab meetings and publication, in progress

Chemical Data Analysis Intern Tacoma, WA
Center For Urban Waters 01/22 - 09/22

- Analyzed Mass Spectrometry data with open source software to verify and flag 150+ potentially environmentally hazardous chemicals from estuary and waste water samples
- Synthesized findings through Excel and Powerpoint for King County and coordinating research labs use

PROJECTS

HiC Database for Kohwi Lab | Final Project for BF768 03/25 - 05/25

- Developed a website utilizing HTML and Javascript elements, interfacing with the SQL database and file server through FLASK and AJAX
- Prioritized the Kohwi lab needs by implementing an interactive UCSD genome browser tab, secure log in, and query flexibility

HCT risk score ML model | Kaggle Competition 01/25 - 03/25

- Developed a cox survival model using Scikit-Survival to predict risk scores for patients undergoing Allogeneic Hematopoietic Cell Transplantation (HCT)
- Cleaned and preprocessed patient data by handling missing values, performing feature selection, and engineering relevant features, achieving a concordance index score of 0.658

Allen Institute for Brain Science | Bioengineering Capstone 01/24 - 06/24

- Optimized an automated fluidics and imaging system for in-situ sequencing of brain tissue by updating GUI allowing users to personalize protocols according to specific research needs
- Implemented secure and seamless parallel upload of experimental data with AWS